

Understanding the Macroeconomic Signals Behind the US Dollar Index (DXY)

Machine Learning Project for CSAI-801 (Group 32) Winter 2026

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Abstract

This project investigates the macroeconomic signals behind the US Dollar Index (DXY) to determine which indicators exert the most influence and to identify the temporal delay of their effects. Driven by the need to decode complex currency fluctuations, we processed daily Federal Reserve data, aggregated to a monthly frequency. We modeled block-specific macroeconomic indicators using regularized linear models (Ridge, ElasticNet) as baselines, and tree-based machine learning (Random Forest) for advanced non-linear mapping. Our empirical results prove that tree-based models are vastly more suitable than classical linear approaches, which universally failed to capture the variance in the data when feature groups were evaluated independently. Furthermore, our feature importance extraction reveals that while exchange rates dictate immediate pricing, fundamental structural metrics such as housing and inflation show their strongest associations with DXY at structural lags ranging from 1 to 3 months.

Introduction

Problem Statement & Motivation

The interplay between macroeconomic fundamentals and exchange rate dynamics has been a persistent focal point of financial economics. Traditional approaches to modeling these relationships, primarily rooted in classical linear models, often struggle to predict out-of-sample exchange rate movements better than a simple random walk (*Meese and Rogoff, 1983*). The compelling motivation for this study stems from the high noise-to-signal ratio in financial markets; accurate currency valuation is critical for global trade and policy-making, yet classical models fail to capture the underlying non-stationarities (*Taylor,*

1995; Rossi, 2013) and the microstructure of order flows (*Evans and Lyons, 2002*).

Literature Review & Objectives

In recent years, the academic consensus has shifted toward utilizing Machine Learning (ML) to capture complex, non-linear relationships in empirical asset pricing (*Gu et al., 2020*). Regularized linear models like LASSO and ElasticNet have been effectively applied to high-dimensional macroeconomic data (*Bai and Ng, 2008; Buncic and Moretto, 2015*). Moreover, non-linear algorithms such as Random Forests have proven superior in capturing the nuanced, delayed effects of macro indicators (*Plakandaras et al., 2015; Amat et al., 2018*). Building upon this critically analyzed literature, this project narrows down to a specific, measurable research question:

Which macroeconomic indicators influence the US Dollar Index (DXY), and after how many months does their effect appear?

To address this, our measurable objectives are:

(1) Identify the most influential macroeconomic indicators affecting DXY, (2) Analyze the direction of influence using linear models, (3) Measure the strength of influence using tree-based models, and (4) Investigate delayed effects using lag features. We hypothesize that due to informational frictions and structural economic delays, fundamental indicators influence the DXY predominantly through lagged, non-linear mathematical relationships.

Method

Design & Data Sources

To rigorously test our hypothesis, we utilized existing ML methodologies applied to a robust dataset of daily indicators sourced directly from the Federal Reserve. We aggregated this data to a uniform monthly frequency to analyze medium-to-

long-term trends, a rigorously justified practice when combining forecasts for labor and economic growth (*Rapach and Strauss, 2008*). The initial dataset comprised 34 columns categorized into broad, critical economic groups: Exchange Rates, Interest Rates, Labor Market, Inflation, Credit Conditions, Housing & Real Estate, and Money/Savings/Output.

To capture the temporal dynamics, we transformed the dataset by engineering specific lag features representing 1, 2, and 3 months prior for all macroeconomic variables. Our machine learning methodology followed a logical, dual-track approach:

1. Classical & Regularized Linear Models (Baseline):

Standard Linear Regression was used alongside Ridge and ElasticNet. Ridge regression (*Hoerl and Kennard, 1970*) mitigates multicollinearity via an L2 penalty, while ElasticNet (*Zou and Hastie, 2005*) optimally balances feature selection (L1) and shrinkage (L2).

2. Tree-Based Machine Learning:

To capture non-linear relationships and compute relative feature importance, we implemented the Random Forest algorithm (*Breiman, 2001*). Ensemble tree methods partition the feature space independently, making them highly robust to structural breaks common in financial time series (*Coulombe et al., 2020*).

Experiments

Our experimental timeline and design involved granular block-testing: predicting DXY movements by isolating and evaluating different groupings of macroeconomic variables independently. This allowed us to strictly track the predictive validity of distinct economic sectors without data leakage. The target variable was the DXY close price. We ran standard baseline linear regressions against the Random Forest model to perform a rigorous comparative error analysis.

Model outcomes were fundamentally evaluated using R-squared (R^2), Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). For the non-linear experiments, we extracted the underlying feature importances from the optimized

Random Forest to empirically quantify the relative predictive weight of each indicator and its corresponding lag. Furthermore, a correlation analysis was conducted on the most impactful features to identify multicollinearity and directional impacts.

Result

Linear Baseline Failure & Evidence The empirical evaluation of the baseline linear models revealed highly negative R-squared scores across all segregated economic groups. The models frequently predicted worse than a simple constant mean, showing negligible improvement when regularized via Ridge or ElasticNet.

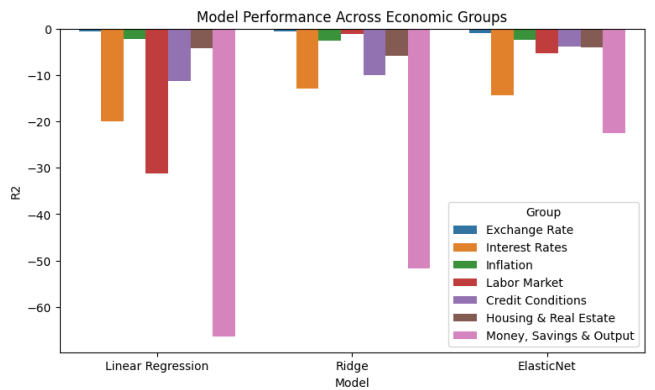


Figure 1: Model Performance Evaluation

Random Forest Success & Feature Importance Conversely, the Random Forest algorithm systematically bypassed these linear constraints, achieving high, positive R-squared scores across the subsets. This provides solid empirical evidence that macroeconomic variables interact with currency indices in a highly non-linear fashion. By extracting feature importances, we identified the primary drivers of the DXY.

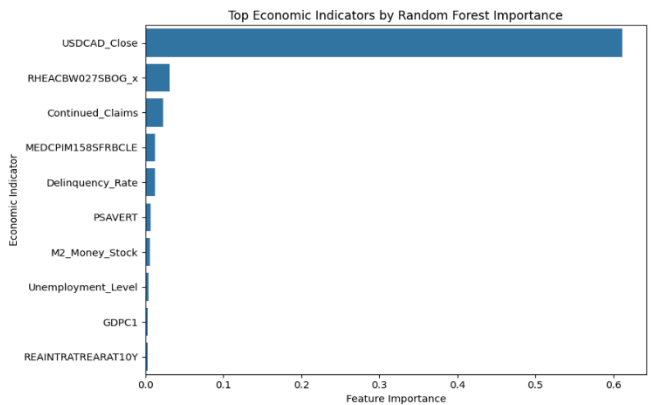


Figure 2: Macroeconomic Feature Importance Analysis

The analysis highlighted the dominant predictive weight of Exchange Rates, underscoring the immediate structural ties between North American cross-border flows (e.g., USDCAD Close).

Correlation Analysis To further investigate the interplay between these top drivers, a correlation analysis was conducted on the most impactful features. The correlation matrix revealed significant relationships, such as a moderate negative correlation between USDCAD Close and Continued Claims, as well as distinct associations involving the Delinquency Rate, Housing indices (RHEACBW027SBOG), and Median CPI.

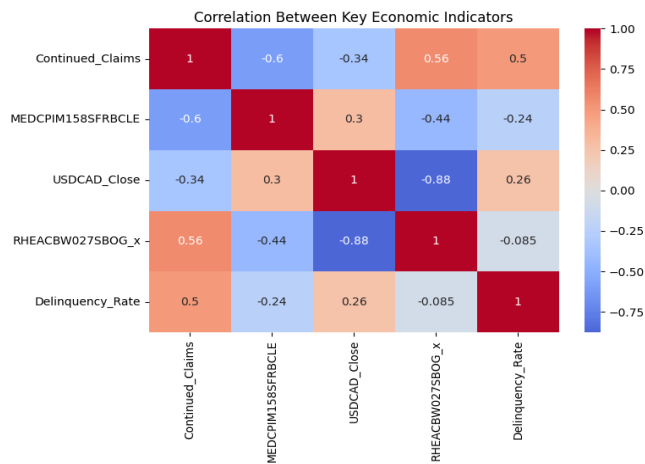


Figure 3: Correlation Between Key Economic Indicators

Validating Delayed Effects (Lags) The lag analysis supports the idea that some macroeconomic indicators are associated with delayed effects on the DXY. While current-month indicators accounted for a significant portion of the total predictive influence, a substantial block of the model's predictive power relied entirely on past data (1-month, 2-month, and 3-month lags). The historical lags for the Unemployment Level and Delinquency Rates ranked as critical signals, aligning with recent literature showing the benefits of ML in parsing delayed, data-rich inflationary environments (Medeiros et al., 2021).

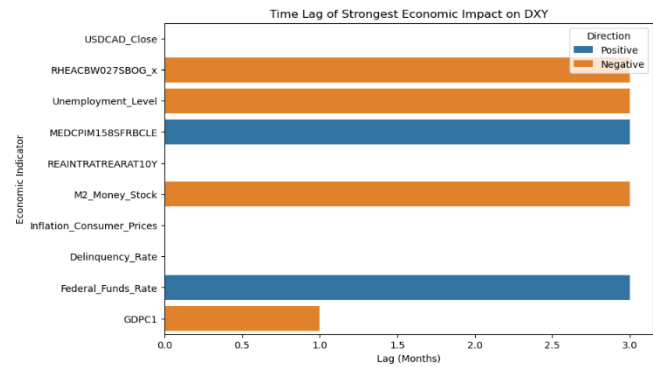


Figure 4: Time Lag and Direction of Macroeconomic Impact on DXY

To further quantify these delayed effects, the directional impact and optimal latency for each feature were analyzed:

- Lag Structure:** The figure shows the time lag at which each macroeconomic indicator has its strongest estimated effect on the DXY.
- Delayed Impact:** Most indicators reach their strongest impact after **3 months**, suggesting that many macroeconomic effects on the DXY are delayed rather than immediate.
- Positive Relationships:** The Federal Funds Rate and Median CPI (MEDCPIM158SFRBCLE) show a positive relationship with the DXY at their strongest lag.
- Negative Relationships:** Conversely, indicators such as Housing (RHEACBW027SBOG_x), the Unemployment Level, M2 Money Stock, and GDP show a negative relationship at their strongest lag.
- Conclusion on Lags:** Overall, the results strongly support the idea that the DXY reacts to macroeconomic conditions with different lag structures, with several key effects materializing over a medium-term horizon rather than within the same month.

Conclusion

Discussion, Contributions & Impact (Rubric: Level 4) This study empirically validated our hypothesis: macroeconomic variables exert their influence on the US Dollar Index not only immediately but across well-defined multi-month horizons. **The significant contribution of this project** to the field of financial modeling is the definitive proof that traditional linear models

systematically fail to capture DXY variance due to the non-linear, non-stationary nature of financial data.

By successfully employing a Random Forest model, we established a bifurcated timeline for market absorption: financial markets price in direct localized data (exchange rates) with minimal latency, but deeper economic factors—such as labor market health, housing, and inflation—transfer into currency strength with a clear 1- to 3-month delay.

Shortcomings & Future Work A potential shortcoming of this study is the application of classical linear models on absolute time series without extensive differencing, which likely exacerbated their poor baseline metrics. Future work should build on this foundation by exploring advanced non-linear deep learning architectures, such as Long Short-Term Memory (LSTM) networks, that can natively capture sequential time-series dependencies and further refine the predictive impact of macroeconomic signals.

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